Algebra 6 — Proportion with a constant k · Laws of indices

Calculator allowed except in question 6. Marks in brackets. ★ = harder question.

1 y is directly proportional to x. When $x = 8$, $y = 12$. [3]

a Find a formula for y in terms of x. **b** Find y when $x = 10$.

c Find x when $y = 4.5$.

> **a** $y = kx$: $12 = 8k \Rightarrow k = 1.5$, so $y = 1.5x$. **b** $y = 1.5 \times 10 = 15$. **c** $4.5 = 1.5x \Rightarrow x = 3$.

2 y is directly proportional to the square of x. When $x = 3$, $y = 45$. [4]

a Find a formula for y in terms of x. **b** Find y when $x = 4$.

c Find the positive value of x when $y = 125$.

> **a** $y = kx^2$: $45 = 9k \Rightarrow k = 5$, so $y = 5x^2$. **b** $y = 5 \times 16 = 80$. **c** $125 = 5x^2 \Rightarrow x^2 = 25 \Rightarrow x = 5$.

3 y is directly proportional to $\sqrt{x}$. When $x = 9$, $y = 6$. [4]

a Find a formula for y in terms of x. **b** Find y when $x = 25$.

c Find x when $y = 1$.

> **a** $y = k\sqrt{x}$: $6 = 3k \Rightarrow k = 2$, so $y = 2\sqrt{x}$. **b** $y = 2 \times 5 = 10$. **c** $1 = 2\sqrt{x} \Rightarrow \sqrt{x} = \frac{1}{2} \Rightarrow x = \frac{1}{4}$.

4 The brightness I of a lamp, measured at a distance d metres from it, is inversely proportional to the square of the distance. At 2 m the brightness is 36 units. [6]

a Find a formula for I in terms of d. **b** Find the brightness at 6 m.

c At what distance is the brightness 9 units?

d The distance is doubled. What happens to the brightness? Explain.

> **a** $I = \dfrac{k}{d^2}$: $36 = \dfrac{k}{4} \Rightarrow k = 144$, so $I = \dfrac{144}{d^2}$. **b** $I = \dfrac{144}{36} = 4$ units. **c** $9 = \dfrac{144}{d^2} \Rightarrow d^2 = 16 \Rightarrow d = 4$ m.
>
> **d** Replacing d by $2d$ gives $\dfrac{144}{(2d)^2} = \dfrac{144}{4d^2} = \dfrac{1}{4} \times \dfrac{144}{d^2}$: the brightness is divided by 4.

5 A fixed amount of gas is squeezed in a cylinder. Its pressure P (in kPa) is inversely proportional to its volume V (in cm³). When $V = 240$, $P = 125$. [4]

a Find a formula for P in terms of V. **b** Find P when $V = 150$.

c Find V when $P = 400$.

a $P = \frac{k}{V}$: $125 = \frac{k}{240} \Rightarrow k = 30\,000$, so $P = \frac{30\,000}{V}$. **b** $P = 30\,000 \div 150 = 200$ kPa. **c** $V = 30\,000 \div 400 = 75$ cm³.

6 Simplify, without a calculator. [4]

a $2a^{-3} \times 5a^{5}$ **b** $\left(c^{-6}\right)^{\frac{1}{3}}$ **c** $d^{\frac{3}{2}} \times \left(d^{\frac{1}{2}}\right)^{-3} \div d^{-1}$

d Find x if $\left(\frac{27}{8}\right)^{x} = \frac{4}{9}$.

a $2 \times 5 \times a^{-3+5} = 10a^{2}$. **b** $c^{-6\times\frac{1}{3}} = c^{-2} = \frac{1}{c^{2}}$. **c** $d^{\frac{3}{2}} \times d^{-\frac{3}{2}} \div d^{-1} = d^{0-(-1)} = d$.

d $\frac{27}{8} = \left(\frac{3}{2}\right)^{3}$ and $\frac{4}{9} = \left(\frac{2}{3}\right)^{2} = \left(\frac{3}{2}\right)^{-2}$, so $3x = -2 \Rightarrow x = -\frac{2}{3}$.

Total: 25 marks

Check: "y varies as x^2" means $y = kx^2$ — always find k first from the given pair · inverse: $y = \frac{k}{x}$, $\frac{k}{x^2}$, $\frac{k}{\sqrt{x}}$ · multiply → add indices, divide → subtract, power of a power → multiply · $a^0 = 1$, $a^{-n} = \frac{1}{a^n}$, $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$.